

Numpy

Python Maths

NumPy

NumPy is a Python library.

NumPy is used for working with arrays.

It also has functions for working in domain of linear algebra, fourier transform, and matrices.

NumPy was created in 2005 by Travis Oliphant. It is an open source project and you can use it freely.

NumPy is short for "Numerical Python".

Uses

In Python we have lists that serve the purpose of arrays, but they are slow to process.

NumPy aims to provide an array object that is up to 50x faster than traditional Python lists.

The array object in NumPy is called ndarray, it provides a lot of supporting functions that make working with ndarray very easy.

Arrays are very frequently used in data science, where speed and resources are very important.

Installation of NumPy

If you have [Python](#) and [PIP](#) already installed on a system, then installation of NumPy is very easy.

Install it using this command:

- `C:\Users\Your Name>pip install numpy`

Import NumPy

- `import numpy`

```
import numpy as np  
  
print(np.__version__)
```

Numpy version

NumPy Creating Arrays

Create a NumPy ndarray Object

NumPy is used to work with arrays. The array object in NumPy is called ndarray.

We can create a NumPy ndarray object by using the `array()` function.

```
import numpy as np
```

```
arr = np.array([1, 2, 3, 4, 5])
```

```
print(arr)
```

```
print(type(arr))
```


Access Array Elements



Array indexing is the same as accessing an array element.



You can access an array element by referring to its index number.



The indexes in NumPy arrays start with 0, meaning that the first element has index 0, and the second has index 1 etc.

- `import numpy as np`

```
arr = np.array([1, 2, 3, 4])
```

```
print(arr[0])
```

Slicing arrays

- Slicing in python means taking elements from one given index to another given index.
- We pass slice instead of index like this: `[start:end]`.
- We can also define the step, like this: `[start:end:step]`.
- If we don't pass start its considered 0
- If we don't pass end its considered length of array in that dimension
- If we don't pass step its considered 1

- `import numpy as np`

```
arr = np.array([1, 2, 3, 4, 5, 6, 7])
```

```
print(arr[1:5])
```

```
print(arr[4:])
```

Data type

- `import numpy as np`
`arr = np.array([1, 2, 3, 4])`
`print(arr.dtype)`

Array Shape

- `import numpy as np`

```
arr = np.array([[1, 2, 3, 4], [5, 6, 7, 8]])
```

```
print(arr.shape)
```

Reshaping arrays

Reshaping means changing the shape of an array.

The shape of an array is the number of elements in each dimension.

By reshaping we can add or remove dimensions or change number of elements in each dimension.

- `import numpy as np`

```
arr =
```

```
np.array([1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12])
```

```
newarr = arr.reshape(4, 3)
```

```
print(newarr)
```


Iterating Arrays

- `import numpy as np`

```
arr = np.array([1, 2, 3])
```

```
for x in arr:  
    print(x)
```

Sorting Arrays



Sorting means putting elements in an ordered sequence.



Ordered sequence is any sequence that has an order corresponding to elements, like numeric or alphabetical, ascending or descending.



The NumPy ndarray object has a function called `sort()`, that will sort a specified array.

- `import numpy as np`

```
arr = np.array([3, 2, 0, 1])
```

```
print(np.sort(arr))
```

Visualize Distributions With Seaborn

Seaborn is a library that uses Matplotlib underneath to plot graphs. It will be used to visualize random distributions.

```
C:\Users\Your Name>pip install  
seaborn
```

```
import seaborn as sns
```

```
import seaborn as sns
import matplotlib.pyplot as plt

# Data
tips = sns.load_dataset('tips')

# Create the scatter plot
sns.scatterplot(x='total_bill', y='tip', data=tips)

# Add labels and title
plt.xlabel('Total Bill')
plt.ylabel('Tip')
plt.title('Scatter Plot of Tips')

# Show the plot
plt.show()
```

pairplot

- A pairplot is a great way to visualize relationships between multiple variables in a dataset. It's commonly used in exploratory data analysis to understand the pairwise relationships between variables. You can create a pairplot using the seaborn library in Python.

```
import seaborn as sns  
import matplotlib.pyplot as plt
```

```
# Load a sample dataset  
iris = sns.load_dataset('iris')
```

```
# Create the pairplot  
sns.pairplot(iris, hue='species')
```

```
# Show the plot  
plt.show()
```

Resources

<https://www.w3schools.com/python/numpy/>

<https://numpy.org/learn/>